§15 · The Living Brain — from equations to a life

A new chapter for "The Learning Brain as a system of equations." §14 turned the model on for four static experiments. This chapter is the living system: one standalone brain, born from a frozen teacher, that then thinks continuously, learns by talking to a stronger model and by seeing the web, on a wake/sleep rhythm it runs itself, growing over a lifetime, with a sense of time — inside fixed compute and memory budgets, watchable and interactive. It is built to be faithful FIRST: five different coupled structures, exactly as §1–§5 describe, not one monolithic network — four of them growable spiking networks (cortex, cerebellum, hippocampus, and the basal-ganglia medium-spiny actor-critic), bound by a §5 neuromodulatory glue (a diffuse scalar tone that gates their plasticity — modulator, not network). Every mechanism is the field guide's; this chapter is the implementation and the honest ledger.

Code: `sapience/brain/*.py`, `run_life.py`, `interface/{dashboard,tui}.py`. Runs on CPU or GPU.

15.1 Five spiking modules, not one network

The guide is emphatic (§11) that the brain is five *different* structures coupled by activation but cut off from each other's gradients (§6). The implementation honours that — each module is its own structure on a shared LIF substrate (`brain/spiking.py`), and each uses the architecture that best mimics *its* region. Four are genuinely spiking and growable; §5 is the neuromodulatory glue — a scalar tone, not a network:

module	structure (region-specific)	teaching signal	file
§3 **Cortex** (spiking, growable)	stacked leaky-integrate-and-fire recurrent layers; **readout taps the analog membrane** of the top layer, not the binary spike	next-byte error via **e-prop** (forward-in-time eligibility + top-down signal + 3-factor gate; §15.17); BPTT (= $\beta \to 0$ PC) opt-in	`spiking_brain.py`
§1 **Cerebellum** (spiking, growable)	sparse LIF granule expansion → Purkinje readout, with a **Golgi feedback-inhibition loop** holding granule sparsity at target	climbing-fibre error, delta rule $\Delta W = -\eta \cdot c \cdot g$	`spiking_modules.py`
§2 **Basal ganglia** (spiking, growable)	LIF **medium-spiny neurons** → critic + softmax actor; the dopamine RPE trains *both* (policy gradient)	dopamine RPE $\delta = r + \gamma V' - V$	`spiking_modules.py`
§4 **Hippocampus** (spiking, growable)	sparse dentate-gyrus separation + a modern-Hopfield store with **spiking CA3 winner-take-all** recall (LIF memory neurons + lateral inhibition → CA1 read-out)	one-shot, novelty-gated write; pattern completion	`spiking_modules.py`
§5 **Neuromod. glue** (scalar)	ACh/NE/DA/5HT tone as the three-factor gate M(t); the **ACh tone scales the cortex learning rate** per phase	$\Delta w = \eta \cdot M(t) \cdot e$	`spiking_modules.py`

The cortex is the deep learner that carries language; the cerebellum corrects, the hippocampus stores/recalls, the basal ganglia supplies reward/curiosity, the neuromodulators gate plasticity and set the wake/sleep tone (§7.2). Spikes and reward scalars cross the boundaries; gradients do not (§6). Unlike an earlier version where these four were built but **orphaned** (only the cortex ran), they are now wired into the living loop (§15.14) so the brain actually *operates* as five systems.

15.2 The spiking substrate

`brain/spiking.py`. Leaky integrate-and-fire neurons: `v ← β·v·(1−s) + Wx + Rs`, fire when `v > θ`. The Heaviside spike is non-differentiable, so the *pseudo-derivative* used by both learning rules is a **surrogate gradient** (fast-sigmoid). By default the cortex learns by **e-prop** (§15.17) — forward-in-time, local, no backward pass; the opt-in `learn_rule="bptt"` reference is surrogate-BPTT, which §3.5 identifies as predictive coding in the $\beta \to 0$ limit, so even the non-plausible reference stays inside the paper's framework while the architecture is genuinely spiking (membranes, thresholds, spikes). The cortex's `TwoCompartmentLIF` implements §3.7 directly: a somatic compartment (basal feedforward +

recurrent drive) and an apical dendrite carrying the top-down error that modulates firing — the substrate the §15.17 dendritic/burst-error toggle uses.

Membrane readout (a measured 2× win). The original cortex applied its readout head to the binary *spike* vector — ~1 bit/neuron, discarding the analog membrane the neuron actually integrates. A controlled A/B (wikitext-2, identical seed/data/budget) showed that reading the graded top-layer **membrane** instead nearly doubles next-byte accuracy (0.17 → 0.34) and drops bits/byte 4.48 → 3.39 — at **zero** faithfulness cost, because the internal computation (inter-layer signals, recurrence) stays entirely spike-based; only the final observation taps the graded population code (as downstream cortex reads rates/LFP, not single spikes). This is now the default. A spike/tanh(membrane) *mix* and a squashed membrane were also tried and are no better; the plain membrane wins.

Two documented, off-by-default cortex levers. (i) `ALIFCell` — an adaptive-threshold LIF (spike-frequency adaptation, the LSNN long-timescale working memory). It is built and correct, but a four-arm A/B found it *loses* at the short byte-context this model uses (seq 48–128), where there is no long-range dependency for its slow state to exploit; it is kept for future long-horizon tasks. (ii) The §3.7 two-compartment top-down apical: faithful hierarchical top-down prediction requires stepping time-outer (layer-within-timestep), which would forfeit the vectorised speed-up below; since the membrane readout already supplies the graded-potential benefit, the two-compartment cell remains in the substrate for the §3.7 exposition but is not the running default. Both are honest trade-offs, not omissions.

A ~7× speed-up, bit-identical. The per-timestep Python loop originally recomputed the input projection `Win(x)` and the readout head every step. Because a feedforward LIF stack has no within-timestep top-down, the layers can run one-at-a-time over the whole sequence (`LIFCell.run_seq`): the input projection is one matmul over all T, the head is one matmul over all T, and only the genuinely-recurrent `Wrec(s)` stays in the loop. This is *mathematically identical* to stepping (verified max-diff 0.0) and cuts a training call from ~1.6–3.4 s to ~0.24 s. Growth stays identity-preserving under it.

15.3 Development: fixed neurons, evolving synapses (§10)

Development follows the biology: **the neuron count is largely set at birth** (neurogenesis is ~complete), so it is a *fixed, settable population* — small for a laptop, or scaled up toward the 100k–1M range the field guide imagines. What **evolves over the lifetime is the synaptic connectome**. The brain is born with a sparse connectome (a settable `syn_density`, default 0.5 of possible connections active); each night of the **child** phase runs **synaptogenesis** — `grow_synapses` activates a fraction of the currently-silent connections with fresh small weights, densifying the wiring — and each night of the **adolescent** phase runs **synaptic pruning** — `prune` silences the weakest active connections, held at zero by a persistent mask so the pruning sticks. The **adult** does slow turnover. So the *synapse count* rises then settles while the *neuron count stays fixed*, exactly as synaptic density peaks in childhood and is pruned in adolescence — all riding the critical-period learning-rate envelope $\eta(t)=\eta\_max/(1+t/t\_c)$ (§10.4).

Neurons can still be grown *deliberately* — `grow_neurons(add)` inserts LIF units whose *outgoing* weights are zero, so the network computes the same function the instant capacity appears (verified: post-grow bits/byte identical to pre-grow), and the synapse mask is padded identity-safely. This is the explicit lever (via `/api/arch`, or a large developmental step) to enlarge the population when there is real experience to justify it — capped at `max_model_gb`. Every part of the architecture (cortex, cerebellum, basal ganglia, hippocampus) reports its live **neuron and synapse counts** (`/api/arch`), and all of it — grow/prune synapses, grow neurons, re-seed a connectome — is editable live per part by API.

15.4 The capability fork, chosen honestly

The design study established a hard truth: a purely local-rule model over raw bytes learns byte statistics, not language. Two faithful ways forward, both grounded in §3.5: - **Spiking, five-module, growable (chosen — fidelity first):** the architecture above, learning by default via **faithful e-prop** (§15.17). Five distinct structures, four genuinely spiking + growable (cortex, cerebellum, hippocampus, basal ganglia) plus the §5 neuromodulatory glue. With the membrane readout and a longer context window its capability has improved from the original ~4 bits/byte to **~3.1–3.2 bits/byte** (born at ~3.3, evolving down over its first nights), with real word-like fragments emerging (*"…was from the … to the … and …"*). Still modest versus a rate GRU — the accepted cost of fidelity — but markedly better than before, and it keeps improving as it lives. - **Rate recurrent cortex (kept as the capability reference):** a byte GRU trained by plain backprop (= β→0 PC). It reaches ~0.5 bits/byte and writes coherent sentences (*"The ocean is … He also added with the starter Cullen …"*), but it is not spiking and cannot grow cleanly. Available as `--core rnn` for when capability is the priority. `brain/rnn_brain.py`.

The default is the spiking core; the fork is a one-line switch, and the paper states plainly which fidelity/fluency point each occupies.

15.5 Birth — the distillation kick-in

`life.birth`. Before it lives, the cortex is trained to competence on a **frozen Qwen3.5-0.8B** teacher's generated outputs (`brain/llm_teacher.py`) plus real text (wikitext-2) — the innate bootstrap. (This birth teacher is distinct from the in-*life* teacher it later converses with, Claude Sonnet 5 via `brain/partner.py`; the UI even labels the two, `teacher_name` flipping Qwen→Sonnet at birth's end.) It is *born able to write* (the rate cortex reaches coherent English at birth; the spiking cortex reaches its modest ceiling faster), then the life evolves it. This front-loads the "fluency needs wall-time" cost into a one-time developmental window, as a genome + rapid early learning do. Birth stops adaptively — on the target bits/byte, OR on a **learning plateau** (no >0.01 bits/byte gain for six epochs), OR a wall-clock cap — because the spiking core floors near ~3 bits/byte and never reaches the rate-cortex target, so without the plateau stop it would burn every epoch; this cut birth from ~220 s to ~150 s.

15.6 One code for every sense

`brain/senses.py`. Sight, language, time and the brain's own voice become one stream of byte-levels 0–255 ("electricity"), each frame tagged by a "nerve" marker — the unified sensorimotor code. Motor output (`brain/motor.py`) is the same code in reverse: writing/ speaking = emitting bytes decoded to text; navigating = emitting a command. Vision enters via a headless browser (`brain/visual_web.py`) that navigates, screenshots and SCROLLS ("reads by scrolling"); its *readable text* is what the language cortex learns, its screenshot is shown as ASCII — raw pixels are not force-fed into the language mind-state (that would be noise).

15.7 The internal clock (a sense of time)

`senses.Clock`. A pacemaker-accumulator plus banks of oscillators at many periods (the striatal beat-frequency model + circadian rhythm) encodes elapsed time to byte-levels and STAMPS every perception — the fastest of §0's clocks, made a sensory channel. The brain always knows what time it is; a fixed time-probe measures how well it has learned to tell it.

15.8 Autonomous wake/sleep, and sleep that learns

`brain/life.py`. Sleep debt (§7.6) rises as the brain learns while awake; the brain DECIDES when to sleep — at least `min_awake`, by `max_awake` at the latest, and in between once the debt crosses a threshold. Sleep is the deep-learning phase, as in biology: it replays the whole-life memory heavily under the default learning rule (faithful e-prop, §15.17; waking thought stays fast while sleep does the consolidation) (§15.10), then applies a §8.4 SHY multiplicative downscale so it also renormalises rather than only potentiating. (Honestly this is an *open-loop* constant downscale — a small fixed factor over all weights — not yet a closed-loop restore of the §9 balance ⟨Δw⟩_wake + ⟨Δw⟩_sleep ≈ 0 scaled to the night's actual potentiation; that, and a protection term for replayed synapses, are named as the next refinement.) Moving deep consolidation into sleep more than doubled the learning rate while keeping waking thought fast (~9 thoughts/s). Each night the brain develops (§15.3).

15.9 Continuous thinking

The brain is never idle: a fast inner-monologue loop generates the next fragment of thought from the persistent spiking mind-state and streams it; a background sense-thread fetches Claude's teaching and web pages without ever pausing the thinking; your typed input is **non-blocking feedback** woven into the same stream (weighted stronger than passive input — more learning steps, tagged "you say:"), with in-band commands (`browse <url>` / `look <url>` redirect vision, `?time` queries the clock); stop and it keeps learning on its own. Every teacher reply is screened first (`_usable_lesson` rejects CLI-error strings and stubs) so the brain never distils *"Error: exceeded budget"* as if it were language — a small but load-bearing robustness guard on the core learning path.

Resonating in parallel. Because the spiking substrate batches over streams, the brain can run *k* thought-streams at once in a single forward pass — `SpikingBrain.resonate(k)` — for essentially the wall-time of one (6 streams measured at 9 ms, the same as one). Every few seconds the living loop resonates, keeps the stream it itself finds most fluent (lowest self-perplexity), and lets that reflection re-enter its mind — parallel exploration collapsing to a better single thought, and it learns from its own best thinking (§9). The teaching conversation with Claude Sonnet 5 runs through `brain/partner.py` (`claude_say` shells the `claude` CLI for a teacher paragraph; `web_topic` pulls clean Wikipedia extracts), verified to feed this same learning path end-to-end: a real lesson measurably lowers the cortex's bits/byte on that lesson. Every teacher reply is screened first (`_usable_lesson`) so a CLI error string is never distilled as language.

15.10 Memory hierarchy — RAM · SSD · eviction

`brain/memory.py`. The whole life of experienced text is tiered within fixed budgets: RAM holds the recent hot buffer; at the RAM/soft limit the oldest hot text is flushed and **compressed** to an SSD segment (zstd-19, ~100×); at the global/hard disk limit the **oldest compressed segments are deleted**. Sleep replays random chunks from hot and a decompressed old segment, so the brain rehearses its whole life. Text compresses enormously, so a long life fits in a small, bounded footprint. Model parameters are bounded by `max_model_gb`.

15.11 Device, precision, scale

`life.pick_device`. Auto CPU/GPU; **bf16 mixed precision (fp32 master weights, matmuls autocast to bf16) on GPU AND CPU** — bf16 *parameters* would wreck Adam's running averages, so the weights stay fp32 and only the compute casts. bf16 on CPU is honoured when asked for, with the honest caveat that without AVX512-BF16 it is *slower* than fp32 (emulated) — it is a GPU-throughput lever, not a CPU one. The checkpoint is device-agnostic (start on CPU, resume on GPU), and — new — **each of the five systems can run on its own device** (`set_part_device` / `POST /api/device`): the living loop converts tensors at every cross-region boundary, so a large cortex can sit on the GPU while a module stays on CPU. Multi-GPU FSDP2 sharding remains the documented next step.

Scaling to a large brain — the sparse cortex. A dense recurrent cortex is O(neurons^2): at 500 000 neurons the recurrent matrix alone is ~750 GB — impossible. Above a threshold each cortical layer therefore switches to a **CSR sparse connectome** — int32 wiring buffers + a dense value `Parameter` + a 1-D active-synapse mask — so memory is O(neurons·fan-in) and the neuron count can reach the hundreds of thousands within RAM. The subtle part is *training*: `torch.sparse.mm`'s own backward transiently densifies a full neurons^2 gradient (measured ~4 GB/layer at 64 k, ~250 GB at 500 k → OOM). A custom autograd (`_SparseConnMM`) computes the value-gradient by a sampled gather/scatter instead — **O(nnz·B), no neurons^2 transient, verified bit-exact against the dense scatter** — so a half-million-neuron brain is genuinely *trainable*, not merely constructible. The small-net **dense path stays byte-identical** and is used below the threshold, so nothing about the default behaviour (or the test suite) changes. All four growable systems are sparse; the cortex is the only one that needs CSR (the modules are neurons×const, i.e. linear, so a masked dense store is already efficient). Honest ceiling: on CPU the recurrent per-timestep `sparse.mm` is poorly threaded, so at 64 k a learn step is tens of seconds — the sparse cortex is built for **GPU**, where that loop is fast; on CPU the trainable-in-real-time sweet spot stays a few-thousand-neuron dense net.

15.12 Checkpoint and resume a life

`life.save_life/load_life`. A life is checkpointed on night boundaries (not per step, which would stall at scale) plus a time fallback: weights + optimiser, and the life state — cycle, age, nights, clock, wake/sleep, mind. `--resume` continues exactly; the compressed episodic archive persists on disk.

15.13 Running it

- `python interface/dashboard.py` — the unified web board + HTTP API (recommended; §15.15). One URL: live charts, thought stream, log, vision, chat, status/config, and launch/stop/kill.
- `python run_life.py` — headless (same life; TensorBoard under `runs/<run>/tb`).
- `python interface/tui.py` — an opencode-style terminal interface. `--core spiking|rnn`, `--checkpoint <run>` to resume.
- `python test/run_tests.py` — the full self-contained test suite (every module + the full life + the dashboard API), 54 tests in seconds, no pytest required.

15.14 The five systems, actually coupled

An earlier version built the four non-cortex modules but never ran them — only the cortex lived. They are now wired into the loop, coupled by **activation and reward only, never gradients** (§6), at *zero* throughput cost (measured 1.00× with the modules on):

- **§2 basal ganglia → curiosity.** A spiking LIF **medium-spiny** actor-critic (growable — its neurons grow alongside the cortex) chooses *which topic* the brain asks Claude about next; its dopamine reward is the **learning progress** on the lesson — the drop in training loss, already computed during learning, so the reward is free. Over time the brain prefers the topics that teach its cortex the most. (Fixing the frozen-actor bug — the old `train_step` updated only the critic — was what made this possible; on a contextual bandit the fixed version learns the optimal policy, 0.33 → 0.98, while the old one stays at chance.)
- **§4 hippocampus → complementary learning systems.** Every experience is fingerprinted; its **novelty** (one minus the best recall match) both **gates whether it is written at all** (familiar patterns are not re-stored, so the buffer keeps the novel ones — encode without overwriting, §7.4) and sets how hard sleep replays it — a novel life consolidates harder. The hippocampus grows its dentate gyrus alongside the cortex each night. The store is a spiking modern-Hopfield associative memory: measured pattern-completion fidelity 1.00 → 0.96 as the number of stored memories goes 10 → 400, where the old covariance-Hopfield version collapsed and could not

even reconstruct the pattern (it returned only a DG code) and had no `grow()` at all — a §10 growability violation now fixed.

- **§5 neuromodulation → plasticity gate.** The ACh/NE/DA/5HT tone is set on every wake↔sleep transition, and — this is the real coupling, not just a readout — the **ACh tone scales the cortex's effective learning rate** on every wake learning step (the three-factor gate M(t) in $\Delta w = \eta \cdot M(t) \cdot e$), so waking encodes at full plasticity and you can throttle it live via `/api/net`. (Only wake and NREM tones are entered; a REM sub-phase is named as roadmap, not claimed as running.)
- **§1 cerebellum → a fast supervised forward model.** Alongside the slow e-prop cortex it runs a *fast* supervised next-byte predictor, learned by its own climbing-fibre delta rule and gradient-cut from the cortex (§6): a bag-of-bytes window is the mossy input, a Golgi-controlled sparse granule expansion the hidden code, a Purkinje readout the prediction. Its error (`cerebellum_mse`) is tracked live. It complements the cortex (a fast vs slow learner, as in the cerebellar–cortical division of labour) and is a monitored forward model, not yet fed back into generation — the honest state of its integration.

On a homogeneous mocked feed the modules are within run-to-run noise on the cortex's own language metric (the mock ignores which topic was chosen, so curiosity cannot differentiate); their measured value is that they are now **free** (no regression) and **faithful** — all five are instantiated and running concurrently, coupled via reward (§2), tone/plasticity-gate (§5), novelty-gated replay (§4) and the cerebellar forward model (§1), where an earlier version had never even instantiated the cerebellum. (What is *not* yet built is the §6 router that would fuse the modules' competing outputs by reliability — the cortex alone drives generation and the cerebellum's prediction is monitored, not fed back; that fusion is named as the next step.) Their real payoff shows on heterogeneous real-world input. Checkpoint/resume persists **all five modules' state and every live-tuned knob**, and a latent resume-after-growth bug (the loader assumed uniform layer width, but growth widens only the top layer) was found and fixed — resume is now byte-identical after arbitrary growth.

15.15 The cockpit — driving and extending the brain by API

A living thing you cannot watch or steer is a black box. The implementation therefore ships a single **web control-plane** (`dashboard.py`) that is also a complete HTTP API — *every* action the browser can take is one documented call (`GET /api/help` lists them), so the brain can be driven entirely programmatically. One command (`python interface/dashboard.py`) serves, on one URL: live metric charts, the timestamped stream of thought, the life log, the ASCII vision, a chat box, a full status/hyperparameter panel, and controls. On a remote host it prints the exact `ssh -L` tunnel to reach it. It supersedes the separate `tail -f` + TensorBoard + TUI (though TensorBoard still logs **~46 metrics** — capability, growth, wake/sleep, the five modules, curiosity, memory, throughput, weight-health — for deep, zoomable history).

- **Lifecycle & compute.** Launch, **graceful stop** (a STOP control file → it checkpoints and exits; no `kill -9`), or force-kill — from the board or the CLI. A run with no checkpoint gets a fresh timestamped folder (old runs untouched); `--checkpoint` continues one in place, and because the checkpoint is device-agnostic you can start on CPU, stop, and resume on GPU or multi-

GPU. An **`auto`** compute mode maximises the hardware (every GPU, else all CPU cores); parallelism (CPU threads, the width *k* of parallel thought-resonance) is a knob. Resume skips re-birth — it is already competent.

- **Everything is live-tunable.** The design principle is that *any* parameter that can change without rebuilding the tensors changes **live**, no restart — the loop reads them each iteration, so a `POST /api/set` propagates on the next tick. That covers the global knobs (budget, wake/sleep windows, perceive-gap, think-chunk, learn-steps, resonance width, threads), the **development / cycle controls** (per-night growth amount, grow/prune ages, and hard toggles to *freeze growth*, *freeze the wake/sleep cycle* so it stays awake, or *freeze learning* so it observes without updating weights), and every **storage cap** below. Only the genuinely structural choices — initial neuron count and layer depth, device, core — need a `POST /api/start` relaunch (which resumes the checkpoint, so nothing is lost). Individual **parts of the net** are tuned through `POST /api/net` (per region, or `target:"all"` at once): the cortex's learning rate / readout mix / context length / sampling temperature, the hippocampus's recall sharpness / sparsity / capacity, the basal ganglia's actor & critic rates, the neuromodulator tones, and **each region's own synaptic growth/prune rate**. (On development: the *neuron count is fixed at birth* — a settable population — and it is the *synapses* that evolve; you set the initial count, the initial synapse density, and per-region grow/prune rates, or freeze it. See §15.3.)
- **Architecture census & live surgery.** `GET /api/arch` reports, per region **and** globally, the neuron count, active-synapse count, parameter count, synapse density, per-layer widths, the per-region grow rate, and the region's device. `POST /api/arch` performs live surgery on any region (or `all`): grow neurons, **set** neurons to a target, grow/prune synapses, **set** synapses to a target count or density, or re-seed a connectome — so the living architecture is both fully observable and directly editable while it runs. `POST /api/device` places any region on its own CPU/GPU (tensors convert at the boundaries).
- **Resource budget, watched against its limits.** `GET /api/resources` (and time-series charts + a usage-bar panel) report RAM (this process **and** the box), VRAM per device, and every bounded store — checkpoint, replay, log, TensorBoard — as *current vs cap*, plus disk free/total. A single `GET /api/diag` is a one-call health check (alive? learning, with recent trends? all five systems firing? + explicit warnings), and `GET /api/value?key=<dotted.path>` fetches *any* value from the live snapshot — all built so the brain can be driven and diagnosed with no eyes on the dashboard at all.
- **Deeper diagnostics (state of the net).** Beyond capability, the board and TensorBoard track the *internal* state: perplexity on what it trains on **and** on what it generates, the entropy of its output distribution, the mean spiking rate (dead-neuron / saturation tripwire), per-layer weight magnitude and spread, the cerebellum's forward-model error, the basal-ganglia policy entropy, and the hippocampus's live recall fidelity — so you can see not just *whether* it is learning but *how healthy each of the five systems is* while it does. And because every live-tuned knob and each module's state is checkpointed, a resumed life continues exactly how you left it — same tuning, same growth schedule, same focus.
- **Bounded everything.** Like the replay buffer (§15.10), all on-disk artifacts stay within live-settable caps that evict the earliest data when full: the life-log truncates to its most recent lines, the TensorBoard event dir deletes its oldest files, the replay buffer its oldest segments, and the

checkpoint is a single overwritten file bounded by the model-size cap — so a long experiment's footprint is fixed and known.

- **Directed teaching & attention.** `POST /api/teach` folds specific content in *now* — raw text, a wiki topic, a web page, or a local **file/dir of code, music notation, or prose** (it learns any byte stream, so its own source code is a valid lesson, verified). `POST /api/focus` redirects the learning feed from broad/random to a target area (topics, urls, mode), so you can say "learn music" or "learn coding" and the curiosity loop curates toward it.
- **Tools & other minds (the unified code, extended).** `POST /api/tools/add` registers any CLI tool or other AI — *opencode* with a free model, a local LLM, a text-to-speech or image command — as a template with an `{input}` slot and an output **modality**. The brain converses with it (autonomously or on demand), and its output is folded into the ONE byte stream by §15.6's encoders: text is learned as language; audio becomes cochlea bytes, an image retina bytes, anything else raw levels. This is the field guide's thesis made operational — *any* sense or tool, or a combination, becomes "electricity" the brain learns from — and it is now extensible at runtime rather than fixed at build time. Each tool carries an optional install command (`POST /api/tools/install`) and enable / manual↔autonomous toggles (`POST /api/tools/toggle`); specs persist with the checkpoint.
- **Replaying what it sensed.** Non-text observations (an audio clip, an image a tool produced) are kept as their encoded byte frames and can be **reconstructed back into playable/viewable media** (`GET /api/observe`) — a deliberately low-fidelity rendering, because it decodes exactly the down-sampled, quantised "electricity" the brain actually received, not the original file. On the board you hear the cochlea bytes and see the retina bytes; in the thought feed each perception is tagged with its modality. It is the closest thing to looking through the brain's own senses.

The point is not the UI; it is that the whole living system — its rhythm, its diet, its senses, the minds it talks to — is inspectable and steerable while it runs, by a human at a glance or by another program through the API.

15.16 The honest ledger

What is real, on CPU: five different structures (§1–§5) — four growable spiking networks (cortex, cerebellum, hippocampus, and the basal-ganglia LIF medium-spiny actor-critic) plus a scalar neuromodulatory glue — coupled by the gradient cut and now **actually operating together** (§15.14), the neuromodulator ACh tone genuinely gating cortical plasticity; a spiking cortex that learns by DEFAULT by **faithful e-prop** (forward-in-time eligibility + a top-down learning signal + the three-factor gate; no BPTT, no weight transport — §15.17), with surrogate-BPTT (= $\beta \to 0$ PC) kept as an opt-in capability reference, reads its graded membrane, runs ~7× faster after a bit-identical vectorisation, and grows by §10 synaptogenesis with the function preserved; parallel thought resonance; a unified byte code for every sense; an internal clock; an autonomous, homeostatically-balanced, learning-during-sleep 24-hour loop with novelty-gated replay and dopamine-driven curiosity; continuous thinking with non-blocking human feedback; a compressed, evicting, whole-life memory in fixed RAM/SSD budgets; checkpoint/resume (byte-identical after growth); born-competent via teacher distillation, then learning from a verified Claude Sonnet 5 conversation; the full §10 developmental arc is real — a **fixed neuron**

population with an **evolving synaptic connectome**: the child runs **synaptogenesis** (activating silent connections, densifying the wiring), the adolescent **prunes** the weakest synapses (mask-persistent), the adult stabilises — with deliberate neuron growth available on demand; and all five systems are genuinely instantiated and running, including the §1 cerebellum as a monitored fast forward model. The whole living system is watchable and drivable through one web board that is also a complete API (§15.15): launch/stop/resume across devices, live hyperparameter tuning, directed teaching, learning-feed steering, and runtime-registered tools / other AIs whose any-modality output folds into the same byte code. The chosen spiking route is faithful FIRST, so its fluency is still modest — improved from ~4 to **~3.1–3.2 bits/byte** by the membrane readout and longer context, with real word-like fragments — with the rate-cortex reference (`--core rnn`, ~0.5 bits/byte, coherent sentences) one switch away when fluency is wanted. Every architecture choice here was settled by a controlled A/B, and the ones that *lost* (adaptive-threshold ALIF at short context, the spike/membrane mix, extra depth) are documented as such, not hidden. Neither route yet has long-range coherence, in-context learning, or reasoning — those exceed local credit assignment and are named, not oversold, as the next chapter: a deeper temporal spiking trunk and the two-compartment top-down path (which needs the slower time-outer schedule). The step §13 called last — instantiate the whole thing and watch it live — is taken here.

15.17 The faithfulness stack, and its measured price

The cortex no longer learns by backprop-in-a-costume. **e-prop is now the default rule** — a forward-in-time approximation to BPTT (Bellec et al. 2020): a per-synapse *eligibility trace* is kept online, a *top-down learning signal* gates it, and a *three-factor* neuromodulator term (the ACh tone, §15.14) opens or closes plasticity. There is no backward pass through time, no weight transport, no separate error network — the update is `@torch.no_grad`, computed by hand from local traces and applied with `w.add_`. Surrogate-BPTT + Adam is kept only as the opt-in, non-plausible **capability reference** (`learn_rule="bptt"`). The single non-local operation that remained (a global gradient-norm clip) is gone: the update is `Δw_ji = −η·M·clamp(mean_t[L_j·e_ji]/N_j, ±Δ)`, where each synapse sees only its own pre-trace, post learning-signal, and the diffuse M, and the per-postsynaptic-**fan-in** normalisation `N_j` (a homeostatic input-scaling) makes the stable rate **width-invariant** — verified identical descent at 8k↔64k↔256k neurons.

On top of that, **every biological constraint is an independent, live-toggleable axis** (extending the `learn_rule` switch — `set_faith(...)` / `POST /api/net`), so each one's cost can be *measured* rather than asserted:

- **Learned feedback (Kolen–Pollack)** vs. fixed-random DFA — the top-down matrix *learns* to align with the readout (its own local gradient + a decay), removing the random-alignment gap without weight transport. Metric: the feedback↔forward cosine.
- **Dale's law** — each neuron is excitatory or inhibitory; its outgoing synapses share one sign (~80/20 E:I), re-projected after every step. Shrinks the usable weight space.
- **Dendritic / burst error** — the learning signal is delivered as an apical **burst** that rides a somatic spike and is thresholded (Naud/Richards) — low-bandwidth, noisy, activity-coupled.
- **Unified two-compartment circuit** — the biological *completion* of the above (see below).

- **Bounded synapses** (Fusi) — weights clamped to ±w_max (real synapses are bounded, low-precision).
- **Firing-rate homeostasis** (metaplasticity) — each neuron's threshold drifts to hold a target rate (Turrigiano), keeping a continually-learning net off silence and saturation.
- **BTSP eligibility** (Bittner–Magee) — the eligibility trace outlives the membrane, widening the temporal-credit window beyond e-prop's ~10-step decay.
- **Differentiated neuromodulation** — the four tones gate four pathways (ACh→cortical encoding/the VIP drive, DA→reward-scaled plasticity, NE→somatic gain, 5-HT→apical patience), not one scalar dial.
- **Stochastic spiking** — probabilistic firing (membrane noise before threshold; noisy vesicle release).
- **Metabolic cost** — a spike-rate penalty in the learning signal (real coding is energy-constrained).

Self-adapting plasticity, and why the hyperparameters are scale-free. The effective learning rate is *not* a dial to hand-tune — it is `eprop_lr_scale · attention`, where **attention** self-regulates on the brain's own learning health: each step's loss is compared to a running baseline, and a loss spike above baseline (a shock, or over-plasticity) *drops* attention → the update shrinks → the representation is protected and re-learns gently (the Yerkes–Dodson arousal→plasticity curve). This makes learning self-healing — an injected weight shock recovers autonomously, and the rate that once caused a runaway stays bounded — and it removes hand-tuning. Sleep, correspondingly, cycles NREM↔REM under the §5 tones with replay depth gated by the day's novelty/debt, so *when* and *how hard* it consolidates is set by the day, not a constant. The design is **scale-invariant by construction**, which is what lets the same hyperparameters move from a 16k to a 256k to a million-neuron cortex without retuning: the base rate is **fan-in-normalized** (÷N_j → the per-neuron drive change, and hence the representation magnitude, is width-invariant — verified: `mem_mag` 1.357 at 16k vs 1.367 at 64k under identical settings), attention runs on the *relative* loss (dimensionless → size-independent), and the Δmax bound is per-synapse. A suite of leading-indicator diagnostics (`mem_mag` = representation magnitude, the true runaway signal that climbs *before* bits/byte does; `update_mag`, `grad_mag`, `attention`, `eff_lr_scale`, `surprise`, `loss_ema`) is exposed in `GET /api/state`, because bits/byte alone lags the dynamics.

One circuit, not separate toggles (the unification). The toggles above are *axes*, but three of them are really one biological circuit, and `two_compartment=True` fuses them: each neuron gets an **apical dendrite** (the §3.7 `TwoCompartmentLIF` compartment, now load-bearing) with its own membrane `ap ← β_ap·ap + gate·(err·B)`; the top-down error is **admitted only through a VIP→SOM disinhibition gate** — SOM inhibition rises with local activity, and VIP, driven by the neuromodulator "learn-now" tone `M`, disinhibits it (`gate = (VIP − SOM)₊`); the apical membrane then **bursts onto somatic spikes** to drive plasticity and **feeds back onto somatic firing** (`drive += g_ap·ap`), with PV supplying fast divisive gain control. So the *substrate* (apical compartment), the *error delivery* (into it), the *interneurons* (SOM/VIP gate), and the *neuromodulator* (drives VIP) stop being four separate features and become **one microcircuit** — the error runs *through* the apical dendrite, not alongside it. This subsumes the standalone `dendritic` toggle (kept for the not-yet-routed comparison).

The measured capability–fidelity curve. Training an otherwise-identical 16k-neuron cortex under each constraint (added one at a time, identical seed/data, 220 steps) gives the cost of each biological commitment — the thing the field usually hand-waves about:

Configuration	bits/byte	cost vs. plausible base
BPTT (non-plausible ceiling)	2.42	−3.78
e-prop + random feedback (DFA)	6.21	+0.01
+ learned feedback (Kolen–Pollack)	6.20	0.00
+ Dale's law	5.58	−0.62
+ dendritic / burst error (standalone)	6.58	+0.38
+ bounded synapses	6.20	0.00
+ firing-rate homeostasis	6.21	+0.01
+ BTSP long eligibility	6.06	−0.14
+ **unified two-compartment** (apical/SOM-VIP/neuromod)	5.90	−0.30
+ differentiated neuromodulation (4 tones → 4 pathways)	5.96	−0.24
+ stochastic spiking	6.19	−0.01
+ metabolic cost	6.34	+0.14
full faithful stack (all mechanisms)	**5.10**	**−1.10**

The honest reading: **the dominant price of plausibility is the learning *rule*** — e-prop costs ~3.8 bits/byte against BPTT — while the individual biological *constraints* are mostly near-neutral or even *helpful* at this scale. Three results stand out. First, **routing the error through the apical compartment beats bolting it on**: the unified `two_compartment` circuit (5.90) is far better than the standalone `dendritic` burst (6.58) — the SOM/VIP-gated apical integration is not only more faithful but more capable than a thresholded side-signal. Second, **differentiating the neuromodulator helps** (−0.24 in isolation): four tones gating four pathways (ACh→encoding, DA→reward-scaling, NE→gain, 5-HT→patience) out-learn one scalar dial, and it couples learning to the wake/sleep cycle (verified: NREM tones down-modulate plasticity vs. wake). Third, **the constraints co-operate rather than compound**: the full stack (5.10) *beats* plain e-prop+learned-feedback (6.20) by ~1.1 bits/byte — the genuinely unexplored result, since stacking that many approximations usually makes a net learn far worse or not at all. Two findings to hold honestly. **Dale's law shows a small *improvement*, not the expected cost** — consistent in sign across seeds (−0.12 to −0.14 at 16k, both seeds; −0.6 in the headline config), a real regularisation effect rather than noise, though its magnitude is config-dependent and it raises the spike rate (so we leave it off the default live run for stability). And **the metabolic penalty correctly *costs* capability** (+0.14) while pulling the spike rate down (0.040→0.033) — the honest energy↔capability trade, and the sign that a first implementation got backwards (a spike penalty must *raise* the per-neuron error, not lower it). (Reproduce:

`runs/fidelity_capability_curve.py`; the numbers refresh into `runs/fidelity_capability_curve.{json,md}`.)

Where this sits, and the permanent gap. On the *learning rule* this is frontier-grade — few groups run e-prop + learned feedback + Dale + dendritic/burst + three-factor neuromod + homeostasis in one spiking net that learns language-like bytes continually. But "faithful to how the brain learns" is roughly a dozen independent axes, and the true cortical algorithm is *unknown* — there is no confirmed single algorithm (predictive coding, the NGRAD/backprop-approximation family, and "prospective configuration"/equilibrium-style relaxation are live, partly-incompatible hypotheses; the strongest phenomenon-level evidence is for prediction-error responses gated by SST/VIP interneurons — Furutachi/Mrsic-Flogel/Hofer, *Nature* 2024 — but that does not uniquely confirm any one algorithm). So the honest status is: frontier on the rule, **mid-field on the other axes, and provably short of "solved," because solved is not yet knowable by anyone.**

The roadmap (named, not oversold). Ranked by whether they genuinely change *learning*:

- *Group A — changes learning (built):* the **PV/SOM/VIP-gated apical circuit** (unified `two_compartment`) and **differentiated four-tone neuromodulation** are now in — the connective tissue between dendritic error and neuromodulation, and the four-pathway modulator that plugs into it. What remains in this tier: **real interneuron populations** — PV/SOM/VIP are currently *functional mean-field* terms (`som = som_b·⟨z⟩`, `vip = ACh`, PV = divide-by-mean), not separate spiking LIF populations with their own connectomes and Dale typing; making them explicit populations is the faithful version. **STDP** (a millisecond spike-timing kernel for the eligibility, not a rate low-pass). **Cascade/complex synapses** (Fusi) for catastrophic-forgetting resistance beyond a bound. A short **relaxation / prospective-configuration** micro-phase (settle activity under top-down nudging before the update) — the biggest single lever per the literature (Song et al., *Nat. Neurosci.* 2024), complementing e-prop's temporal traces with spatial relaxation, and a knob to test a predictive-coding objective against the current global readout.
- *Group B — when it learns:* **sharp-wave-ripple / theta-gamma gating** of replay (we schedule replay by a debt heuristic; biology gates it on oscillation events); **adult dentate-gyrus neurogenesis** (the one place automatic neuron growth is the faithful choice).
- *Group C — realism, steeper diminishing returns:* richer dendrites (NMDA plateaus), stochastic spiking + a metabolic/energy term, short-term plasticity.
- *Group D — the setup, not the synapse:* a closed **sensorimotor / active-inference loop** where the brain's outputs contingently reshape its future input. This is arguably the deepest gap in how the brain learns — biological learning is grounded in an action–perception loop with consequences — and it is about the *setup*, not the rule. Our brain reads and talks to Claude, but its words do not yet change its world.

Faithfulness is **asymptotic**: the brain has effectively unbounded biophysical detail, so there is always another axis. The research value is not checking every box — it is (1) separating the axes that change learning (group A) from the realism that does not (group C), and (2) *measuring the fidelity-vs-capability curve as each is added*. That curve — "here is exactly how much each biological constraint costs" — is the contribution; the full survey lives in `runs/cortical_algorithm_research.md`.

§16 · The deeper brain: one neuro-endocrine-oscillatory control loop (R&D)

§15 gives five spiking modules + a self-adapting cortex, but every cycle fires all five and sleep replays a **raw byte buffer** — two things the brain almost certainly does not do. Three literature-grounded R&D threads (full surveys + citations in `runs/{memory_architecture,drive_stress,dynamics_oscillations}_research.md`; integrated design in `runs/deeper_brain_integrated_design.md`) converge on a single missing controller that sits *above* the existing tones, the self-adapting `attention`, the sleep-debt, and the dopamine critic — coupling only to fields that already exist:

> **endocrine drive/cortisol** (sets the arousal/entropy operating point + **sleep pressure**) → the **ultradian sleep-cycle** (NREM/REM schedule) → **ripple-gated generative consolidation** of the **generalized-in-the-net** memory → gated by **oscillatory dynamic states** (which regions fire, at what frequency, set by attention) → feeding back into the self-adapting attention.

Memory as generalized-in-the-net, not a buffer — the idea, and an honest negative result. The raw replay buffer is biologically wrong: hippocampal replay is *reconstructive/generative* — it builds never-experienced shortcuts (Gupta 2010), pre-plays untraversed paths (Ólafsdóttir 2015), and sweeps to *future* goals (Pfeiffer-Foster 2013); under Complementary Learning Systems (McClelland 1995; Kumaran 2016) the cortex accumulates *generalized* structure in its weights, nothing stores raw episodes. The buffer-free mechanism is **generative self-replay** (Robins 1995; Shin 2017; van de Ven 2020): the cortex *dreams* from its own dynamics and hard-learns the dreams — built here as `SpikingBrain.generative_replay()` (diverse high-temperature dreams from sparse byte-cues, with two anti-"overfitted-brain" safeguards: a veridical-anchor fraction and a held-out acceptance monitor). **But the honest result is negative.** A single-seed pilot looked promising (generative 0.274 > buffer 0.255 > none 0.232); a **3-seed** re-measurement did **not** replicate (retention none 1.131 / buffer 1.031 / generative 0.236), and at anchor-fraction 0 the dreaming *corrupted* the representation — worse than doing nothing. So this is a compelling hypothesis that our current small-cortex implementation does **not** yet deliver; it stays **OFF** and the §16 ledger records the null. Whether a veridical anchor or a larger cortex rescues it is open (`runs/deeper_brain_measure.py`).

The drive/stress layer. A `SpikingEndocrine` of slow scalars — a **drive deficit** (leaky integrator, met by a satiation event → a homeostatic-RL reward, Keramati-Gutkin, fed into the existing dopamine critic), **cortisol** (rises with unmet drive + prediction-error; an inverted-U on plasticity — acute sharpens, chronic impairs; and it *is* the sleep-pressure term), and **mood** (a dopamine EMA that damps cortisol — resilience). "Satiation → focus" is principled: a satisfied drive lowers tonic NE → the phasic, focused mode (Aston-Jones- Cohen). (Energy is modelled as an opportunity-cost bias, not a fuel gate — the glucose/ego-depletion account is discredited, Hagger 2016.)

The dynamics layer. Not-all-regions-active is an **ignition gate** (global workspace, Dehaene) — each system ignites only if its salience beats a threshold set by a single **entropy knob β** (the normal↔psychedelic axis, REBUS/Carhart-Harris). Attention shifts the **processing frequency**

(gamma-fast window when focused, alpha-slow when disengaged, theta sampling) — a variable effective eligibility-window over the fixed tick. Sleep becomes a fixed **ultradian FSM** (SWS-heavy early → REM-heavy late) with SO-spindle-ripple coupling gating *when* consolidation commits.

Honest status + phased plan. An adversarial pass (in the design doc) confirmed the *control* couplings are one real loop but flagged that the current histogram-based hippocampus cannot seed a semantic "gist" — hence P0 uses sparse *byte-cues*, not vector reinstatement. Build order by value-per-effort, each behind a toggle with an acceptance test, landing one piece at a time (the lesson of the faithfulness build):

- **P0 generative self-replay — built; benefit NOT established.** `SpikingBrain.generative_replay` (above). A single-seed pilot suggested it beat the raw buffer on forgetting-resistance; a **3-seed re-measurement did not replicate** (ledger below), and buffer-free dreaming *without* a veridical anchor corrupted the representation. It stays OFF — not by caution but by measurement. Live via `/api/set sleep_mode=generative`.
- **P1 `SpikingEndocrine` — built; neutral on loss, measured behavioural value** (`brain/endocrine.py`). Three bounded hormone scalars — a drive deficit (met by learning-progress → a homeostatic-RL reward into the dopamine critic, and satiation → low NE → focus), cortisol (a **one-sided** gate on plasticity — calm→optimal learning is full, only chronic-high cortisol impairs, capped further by allostatic load; and it is the sleep-pressure term), and mood (a dopamine EMA that damps cortisol). Verified: satiation emits reward + focuses; calm cortisol leaves plasticity full while chronic-high impairs; chronic stress accrues allostatic load, and **sleep recovers it** (`runs/endocrine_test.py`). Live via `/api/net endocrine`; metrics `cortisol/drive/mood/allostatic/plasticity_gain`.
- **P2 dynamic states — built; neutral on loss, ~50% compute saved** (`brain/dynamics.py`). A single entropy knob β (the normal↔psychedelic dial) drives **selective ignition** — auxiliary systems run only when salient, so the brain is not all-on every cycle — and attention sets the **processing frequency** (a gamma-short eligibility window when focused, alpha-long when disengaged). Live via `/api/net dynamics`; metrics `beta/n_active/eff_freq`.

The obligation of breadth: measured, not asserted. §15.17 taught that a mechanism earns its place only after an A/B on the metric it *actually* claims to touch. `runs/deeper_brain_measure.py` discharges that for §16 and writes the numbers to a committed artifact (`runs/deeper_brain_measure.json`): a bits/byte A/B for P1/P2 (identical 16k cortex, same seed/data), a **3-seed** forgetting-resistance test for P0, and *behavioural* A/Bs that bits/byte cannot see. The honest ledger:

§16 mechanism	measured	verdict	where its value is (or isn't)
P0 generative self-replay	forgetting-resistance retention, **3 seeds**: none **1.131** / buffer **1.031** / generative **0.236** (noise ±0.22)	**NO benefit** — does not beat the buffer or no-replay; anchor-free dreaming *corrupts*	unproven; **default OFF** by measurement
P1 endocrine	bpb 3.999 → **3.998 (−0.001)**; **stress-protection** retention on/off **0.751 vs 0.302**; drive→arousal gen-entropy **3.96 vs 3.68**	**NEUTRAL on bpb; measured behavioural value**	protects prior knowledge under stress; drive→arousal→exploration
P2 dynamics	bpb 3.999 → **4.003 (+0.004)**; **~50%** of systems ignite per cycle	**NEUTRAL on bpb; ~50% compute saved**	not-all-on realism + selective compute
adult-DG neurogenesis	separation/recall under interference **1.00** → **1.00**	**no measured gain** (test saturated)	separation of new memories — needs a harder test

The measurement earned its keep **twice**. First it exposed a design flaw: an early run showed the endocrine *hurting* bits/byte (+0.066) because a textbook *bidirectional* cortisol inverted-U throttled **calm** learning; the impairing arm (chronic-high) is the load-bearing one, so the gate is now one-sided and the endocrine is neutral on loss. Second — and this is the point of measuring — it **retracted an overclaim**: P0's single-seed "beats the buffer" did **not** survive three seeds, and dreaming without a veridical anchor made retention *worse than doing nothing*. **The honest verdict: no §16 mechanism is a bits/byte win.** P1 and P2 have *measured non-loss* value — stress-protection and drive-modulation for P1, ~50% compute-selectivity for P2 — so they are richer than cited scalars; P0 and neurogenesis show **no measured benefit yet** and are honestly labelled as such. All default **OFF**, now by evidence rather than caution.

Two faithfulness constraints the same rule already covers, lest they read as "missing": **metabolic cost** (a spike-rate energy penalty) and **stochastic spiking** are built and measured as §15.17 toggles — metabolic costs ≈ +0.14 bpb (the expected price of an energy constraint; the brain learns to spike less), stochastic is ≈ neutral — both honest, both default off.

Adult-DG neurogenesis — built, measured null. The dentate gyrus keeps adding granule cells in adulthood (Aimone/Gage), improving separation of *new* memories; `SpikingHippocampus.grow()` already adds DG cells and re-separates stored patterns identity-preservingly, so it is wired into the adult wake-phase behind a toggle (`/api/set neurogenesis`). But the A/B shows **no measured gain** (recall under interference 1.00 → 1.00 — the test saturated); it stays OFF until a harder separation regime can tell the two apart. Honest row, in the ledger.

- **P3 (roadmap — explicitly NOT built).** Named so the ledger is not mistaken for completeness: the full SO-spindle-ripple sleep FSM (ripple-*gated* consolidation commit); a richer hippocampal trace so reinstatement carries *sequence*, not just letter statistics; typed PV/SOM/VIP interneuron **populations** (today they are functional scalars, not spiking sub-nets); an STDP timing kernel; and an **embodiment / closed sensorimotor loop**. Each will land the §15.17 way — one toggle, one A/B, one honest row — or not at all.

Every §16 mechanism defaults OFF (opt-in, each A/B-measured above — P1/P2 do-no-harm with measured non-loss value; P0 and neurogenesis measured nulls), is a device/dtype-agnostic scalar controller, and persists across checkpoints — so the deeper brain can be switched on and tuned live without disturbing the running cortex.

§17 · The gap-map: the missing fundamentals, built and wired

A rigorous critique named the mechanisms the brain was still missing or carrying only nominally: a closed **sensorimotor loop** (the single largest gap), **predictive coding** as a competing cortical objective, real spiking **interneuron populations** (not mean-field scalars), hippocampal **temporal sequences** (not just content), **STDP** timing, **ripple-gated** consolidation, **short-term plasticity**, dendritic **NMDA plateaus**, **glia/astrocytes**, **neuropeptides**, **laminar** cortical structure, and **grid/place** cells. All twelve are now built as first-class mechanisms and wired into the live architecture under one contract: each is **default OFF**, **live-tunable via `/api/net` with no restart**, **persisted** across checkpoints, **metric'd** to `/api/state`, **device/dtype/FSDP-aware**, **scale-verified at 256k** (no dense O(N^2) state, no width-dependent pathology), **byte-identical to the prior brain when off**, tested, and — critically — **inter- connected** with the existing systems rather than bolted on. The honest status ledger (present ≠ load-bearing; every earns-keep A/B is the standing next step, so all default OFF until each measurably earns its place):

§17 mechanism	what it adds	key interconnection	earns-keep A/B (pending)
Embodiment (`embodiment.py`)	closed obs→BG-actor→ACTION→world→reward→learn loop (active inference) in a GridWorld	world reward → the SAME §5 dopamine tone the cortex e-prop gate reads; endocrine NE/pressure → explore temp; Dyna replay in sleep	navigation return vs random (already: **100 % goal-reach** in-run)
Predictive coding (`predictive_coding.py`)	a THIRD cortical rule `learn_rule='pc'` (Rao-Ballard/Friston, precision-weighted, β→0 instantaneous)	shares e-prop's spmm/sddmm/`_upd`; peer of eprop/bptt	pc vs eprop bits/byte (already: **learns, bpb 7.9→3.9**)
Interneuron populations (`interneurons.py`)	real spiking PV/SOM/VIP LIF pools replacing the mean-field scalars	drop-in inside the two_compartment apical circuit	bits/byte + stability vs scalar
Hippocampal sequences (`theta_seq.py`)	ordered trajectory storage + fwd/reverse replay (theta sequences)	rides the hippo DG; SWR-gated; seeds CLS dreams	ordered vs scrambled next-item
STDP (`stdp.py`)	pair-based spike-timing plasticity, additive to e-prop (mix)	folded into the eligibility grad before the shared `_upd`	does ms timing help a byte LM?
Ripple consolidation (`ripple.py`)	SWR point process gates WHICH sleep-replay commits	gates generative + buffer replay; endocrine pressure scales density	retention gated vs ungated
Short-term plasticity (`synaptic_stp.py`)	Tsodyks-Markram facilitation/depression on transmission	eligibility carries the transmitted z⊙g; NE→release-prob	temporal-processing bits/byte
Dendritic NMDA plateau (`plateau.py`)	regenerative latched apical nonlinearity	extends the two_compartment apical; stretches eligibility→BTSP	delayed-credit task
Glia / astrocytes (`glia.py`)	slow per-neuron astrocytic field, one-sided metaplastic brake	gates the plasticity update + metabolic cost	runaway-stabilisation

§17 mechanism	what it adds	key interconnection	earns-keep A/B (pending)
Neuropeptides (`neuropeptides.py`)	slow OXT/ORX/CRH modulators (companion to the endocrine)	CRH gains cortisol's driver, OXT relieves its pool — one integrator	stress-protection / exploration
Laminar microcircuit (`laminar.py`)	canonical L4/L2-3/L5-6 column via a per-edge adjacency mask	thins the sparse CSR; per-neuron effective fan-in norm	structure vs flat pool at equal neurons
Grid / place cells (`spatial.py`)	entorhinal grid + hippocampal place + path integration	φ for the embodiment nav-BG; SPACE_NERVE proprioception	navigation return vs tabular

Two honest caveats. (1) This is the *reframe that matters*: implementing the parts list is the achievement; demonstrating that the parts, together, are **load-bearing** is the work these A/Bs now enable — which is exactly why the experiment phase can begin (the fundamentals are no longer missing). (2) "All the science" remains an overstatement in principle: e-prop, predictive coding and target-prop are *rival* theories of the same cortical learning (you cannot have all three be true at once — they are offered as selectable `learn_rule`s), and the biophysical tier is unbounded. What is true is: **most of the major systems-level hypotheses are now present, wired, interconnected, honestly measured-or-labelled, and independently switchable.**

§18 · The boundary — what is deliberately *not* simulated, and why

§17 closed the list of missing *learning* mechanisms. This section draws the complementary line: the mechanisms the model deliberately does **not** implement, the principled reason, and why crossing that line would make the system worse **at its own goal**. It belongs in the honest ledger because a model must state its exclusions as plainly as its inclusions — one that never says what it leaves out invites the reader to assume it does everything, and "everything" is precisely the overstatement §17's caveat began to correct.

18.1 Two different fidelities

"Brain model" names two research programmes that are routinely conflated, and this system belongs to exactly one of them.

- **Fidelity-of-substrate** — reproduce the *physical* brain as accurately as the instruments allow, and read cognition out of that physics. This is the biophysical-simulation tradition. **Blue Brain** (EPFL) reconstructs a rat neocortical microcircuit of ~31,000 neurons in which each cell is a morphologically detailed multi-compartment cable with **Hodgkin–Huxley ion channels**, wired by ~37 M synapses with measured receptor kinetics; the **Allen Institute** fits biophysically detailed single-neuron and network models to a cell-type atlas and electrophysiology; **connectome-constrained** simulations run on *measured wiring* — the full adult *Drosophila* connectome (~140k neurons, FlyWire/FlyEM) and *C. elegans* (302 neurons). The tools are **NEURON** and **Arbor** (multi-compartment), **NEST** (point neurons at supercomputer scale), **Brian2**, and neuromorphic silicon (**SpiNNaker**, **BrainScaleS**). On raw biophysical accuracy these are — and will remain — orders of magnitude ahead of anything here. That is their entire purpose.
- **Fidelity-of-learning** — reproduce *how the brain learns and behaves*, at whatever level of substrate abstraction makes that tractable, and judge the model by whether it **learns and acts** the way a brain does. This is the programme of this paper (§15–§17).

The decisive fact separating them: **the fidelity-of-substrate models do not learn or behave.** Blue Brain does not acquire language or navigate; the fly connectome is a wiring diagram, not an agent; even **Spaun** (Eliasmith), the largest *functional* spiking model at 2.5 M neurons, is largely hand-compiled with limited plasticity. High-precision simulations *replay* biophysics; they do not *live*. So the two programmes are not rivals on one scale — they optimise different objectives. "The most precise brain simulation" is a title in the *other* programme, held by Blue Brain and its kin; it is not a target of this one, and not a yardstick this one is short of.

18.2 The inclusion rule

Every mechanism §15–§17 added passes one test, and every biophysical-tier mechanism fails it:

> **Include a mechanism iff it changes what the network can *learn or do* — not merely how physically accurate its substrate is.**

e-prop, STDP, predictive coding, the dendritic-error microcircuit, embodiment — each alters the *computation or the credit assignment*, i.e. the learnable function or the behaviour. Ion channels, dendritic morphology, and receptor kinetics alter the *accuracy of the substrate* while leaving the learnable function unchanged: the LIF abstraction already captures the computable integrate→threshold→reset dynamics (§15.2); the two-compartment split already captures the one dendritic distinction that is computationally load-bearing — the apical top-down error path (§15.17); the NMDA-plateau abstraction (§17) already captures the one regenerative nonlinearity that matters for delayed credit. Their full biophysical form buys precision on an axis the goal does not score.

biophysical mechanism	changes *learning / behaviour*?	verdict
Hodgkin–Huxley ion channels	no — LIF already captures the computable dynamics; HH only adds stiff Na/K/Ca ODEs at a sub-ms timestep	**exclude** (and it would *break e-prop*, which is derived for LIF/ALIF, not an HH cable)
Full multi-compartment morphology	no — the load-bearing soma/apical distinction is already two-compartment; the NMDA plateau is already added	**exclude**
Receptor kinetics (AMPA/NMDA/GABA time constants)	no — STP + plateau + surrogate dynamics already approximate the fast/slow/inhibitory functional split	**exclude**
Extracellular field / ephaptic coupling	no — a negligible, speculative effect on credit assignment	**exclude**
Detailed neurovascular / metabolic coupling	no — the *functional* abstraction (metabolic spike penalty + glial slow brake, §17) is already present	**exclude**
A measured connectome (as an initialisation prior)	*maybe* — connectivity structure (motifs, E/I ratio, laminar/modular graph) can shape the learnable function	**the one exception — worth a single A/B**

18.3 Why crossing the line would make it worse

The exclusion is not merely "unnecessary"; integrating the biophysical tier would actively **regress** the system on its own metric:

- **Scale collapse.** Biophysical detail costs ~10–100× per neuron and forces sub-millisecond integration; the trainable-in-real-time regime would fall from the 10^5–10^6-neuron sparse-CSR range (§15.11) to ~10^3–10^4 — the scale at which biophysical simulators already operate, and *below* the scale at which learning here is interesting.
- **A different simulator.** Multi-compartment HH belongs to NEURON/Arbor, not the vectorised PyTorch e-prop loop; the entire substrate — the forward dynamics of §15.2, the O(nnz) sparse ops of §15.11, the local `_upd` of §15.17 — would have to be discarded to host it.
- **The learning rule breaks.** e-prop's eligibility trace and pseudo-derivative are *defined on the LIF/ALIF membrane*; there is no drop-in e-prop for a full Hodgkin–Huxley cable. The plausible, online, forward-in-time learning that is the entire point (§15.17) does not survive the substrate swap.

The trade, stated plainly, is: surrender a functional, scalable, *learning* agent to obtain a slower, smaller, *non-learning* replica of a substrate whose detail the goal does not use — a strictly worse instrument for the question being asked.

18.4 The honest positioning, and the one experiment worth importing

Stated precisely: this is **not** the most precise brain *simulation* — that title is Blue Brain's and the connectome sims', in a programme this model does not enter. It is, as far as we are aware, the most complete *integration of systems-level learning-and-cognition mechanisms into one continuously-learning, behaving agent* — a claim in the fidelity-of-learning programme, where the biophysical simulators do not compete because they do not learn. The mechanisms it "lacks" are not gaps in its own programme (§17 closed those); they are the constitutive detail of the *other* one.

The single biophysical idea that lives on *this* model's axis — because it plausibly changes the *learnable function* rather than only the substrate — is **connectome-as-initialisation**: seed the sparse recurrent connectome from realistic connectivity *statistics* (or a real graph) instead of random fan-in, and run one earns-keep A/B under §17's contract against the learned-sparse baseline — does structured initial wiring learn faster, or reach a lower bits/byte? That is the one experiment worth importing from the biophysical world. Everything else on that tier is admired in Blue Brain and left there: not because more realism is unwelcome, but because on the axis this model is graded it would not move the score, and it would cost the very properties — scale, plausible online learning, a living loop — that define the work.

§19 · A possible extension — merging a large language model as an amnesiac-oracle organ

The architecture's honest capability ceiling (§15.4, §18) is language fluency and knowledge breadth. The most powerful available source of both is a large pretrained LLM — which this brain already *talks to* as a teacher (`partner.py`) and can register as a tool (`tools.py`, §15.15). This section specifies, in detail, what it would take to go beyond *talking to* an LLM to *merging* one into the living system as an organ — why the merge must be asymmetric, the concrete binding layer (most of whose hard ingredient already exists), the test that separates a merge from a tool call, and the wall that a frozen model cannot cross. It is offered as future work, not a present claim.

19.1 Why tool-use is not a merge

The brain already calls an LLM (Claude Sonnet via `partner.py`; any AI via the tools registry). But that exchange is **text-in / text-out** — a narrow discrete bottleneck — and the model's internal representations and per-call state are discarded. Two minds converse; one mind does not result. Three structural facts make a *symmetric* fusion impossible with a frozen model: (1) its weights are frozen and an API exposes **no gradient access**, so the brain's e-prop (§15.17) cannot update it; (2) it holds **no persistent state** to share a continuity with; (3) its knowledge lives in ~10^{11} parameters that **cannot be poured into** a 10^5–10^6-neuron spiking net (a capacity gap) nor absorbed wholesale by a weak local learner. That wall is constitutive, not incidental — and naming it is what makes the rest of the design honest rather than hopeful.

19.2 The reframe — heterogeneous binding, one seat of continuity

A biological brain is not homogeneous either: it is heterogeneous organs bound into one mind by **shared state** and a **single locus of continuity**. That is the template, and it forces an asymmetry — one component must be the *self*. Here the self is the **sapience brain**: it holds the continuity (the persistent mind-state, §15.9), the whole-life memory (§15.10), and the drives and identity (§16). The LLM becomes a **language-and-knowledge cortex** the self queries *into its own persistent state*. This resolves the obvious objection ("the LLM still forgets"): the LLM organ forgets every call, but the merged *self* does not, because the self is the brain, which remembers — exactly as individual cortical computations are transient while the hippocampus and the persistent cortical state carry the thread. The merged identity is the brain's; the LLM extends it, not the reverse.

19.3 The binding layer — three couplings

The difference between "phone a friend" and "a language organ wired into the cortex" is three couplings — and the hardest ingredient already exists in `llm_teacher.py`, which captures the teacher's **hidden states** (`mid` / `tgt` from the residual stream), not only its tokens.

coupling	tool-use (today)	merge (extension)	current substrate
up — LLM → brain	the reply *text* is learned as language	the residual-stream **hidden state** is projected into the brain as a rich sensory/cortical channel — a *shared latent space* both minds read, not a discrete string	`llm_teacher.features()` already extracts `mid` / `tgt`; needs a learned projection into the cortex input (§15.6)
down — brain → LLM	a fixed teacher prompt	the brain's current **membrane / mind-state** is projected into the LLM's context (a soft-prompt encoder), so it generates *conditioned on what the self is thinking*	new: a brain-state → context encoder
persist — into the self	the reply is consumed and forgotten	every interaction accretes into episodic memory (§15.10) and is **continuously distilled** into the brain's living weights, so the self internalises a *flavour* of the organ's competence over its life	partial: birth distillation exists (§15.5); needs an in-life continuous-distillation loop

- **Latent up, not text up.** Text is a narrow discrete bottleneck; the residual-stream representation is a shared latent. Folding it in as a *sense* (§15.6) turns the exchange from a message into a nerve.
- **Condition down.** Projecting the brain's state into the LLM's context makes it "think the self's thoughts" rather than answer detached queries — the organ's computation is *bound to* the self's

state.

- **Persist and distill.** Continuous distillation makes the merge *deepen with the life* instead of resetting each call. It is **capacity-bounded** by construction — a small spiking net cannot absorb 10^{11} parameters, and e-prop is a weaker optimiser than backprop (§15.17) — so the transfer is a *flavour of competence*, not the whole mind. Stating that bound is what keeps the claim honest.

19.4 The test that separates a merge from a tool

One criterion decides it: does the LLM's contribution **become part of the brain's persistent, evolving self**, *and* does the brain's state **shape the LLM's computation**, such that neither is fully itself without the other in the loop? If the reply is used and forgotten with no lasting change to the unified state — a **tool** (what the registry does today). If it is woven into a continuity that remembers and re-shapes both sides — a **merge** (heterogeneous, as a brain's regions are).

19.5 The hard wall, and the only road through it

The extension above yields an **asymmetric** merge: one living self that has bound a frozen oracle. True **symmetric** fusion — both halves mutually plastic, co-adapting on one shared state, the language half no longer amnesiac — requires an **open-weights LLM trainable in-process**, so the transformer and the spiking system can share state and even gradient-couple. That changes the compute profile entirely and is a large undertaking, but it is the only path past the asymmetry. With a frozen API model the ceiling is explicit, and worth stating plainly: **one continuous mind, seated in the spiking brain, that has bound a powerful but amnesiac language organ into its remembered life** — already a stranger and more capable thing than either component alone, and the version this architecture can actually reach. It is, fittingly, the same design principle as the rest of the system: not one homogeneous network, but distinct structures coupled by shared activation and a single seat of continuity (§15.1) — extended, now, to a structure that was trained in a different world.